LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	36
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose and principle of operation of flap compensating circuit.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose and principle of operation of flap compensating circuit.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of circuit diagrams in IB; Instructor supervision during Q&A; session.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) How does a servo actuator control surface movement?
(c) Why is compensation necessary in flap systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Flap Compensating Circuit • Purpose of Flap Compensation • Basic Circuit Components • Signal Flow Path • Role of AFCS in Compensation	L & Ds	PPT, IB	15 min
(b)	• Operation Principle of Flap Compensating Circuit • Interaction between Flap and Servo Signals • Feedback Mechanism in Circuit • Adjustment of Compensation Levels • Impact on Aircraft Stability	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting Common Circuit Failures • Maintenance Checks for Flap Compensating Circuit • Safety Precautions During Inspection • Documentation Requirements • Integration with Other AFCS Systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the purpose of the flap compensating circuit in the AFCS.
(b) Describe how the circuit ensures proper flap operation during flight.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of the flap compensating circuit apply to the duplex servo actuators discussed in LP 37.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation of Duplex servo actuators.	Refer to the master syllabus reference list	LP 37

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min